Vision Document

Meal Planner App for the Elderly

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Project Overview

The Meal Planner App for the elderly is a digital solution designed to support healthy, safe, and independent living through personalized meal planning. The application aims to recommend weekly meal plans that align with professional nutritional advice while also respecting individual user preferences such as cultural background, taste, allergies, medication constraints, and budget.

The primary goal of the project is to reduce the cognitive and physical effort associated with daily food-related tasks for elderly users.[1]. By providing pre-approved recipes, clear meal preparation guidance, and automatically generated shopping lists, the app seeks to minimize stress around meal planning and improve overall nutritional well-being.

In addition to supporting elderly users, the system is intended to assist caregivers and family members by enabling better coordination around shopping and meal preparation. The application focuses on accessibility, simplicity, and reliability to ensure it is usable by users with varying levels of technical ability.

This project will be developed using an agile methodology[2], allowing the vision and scope to evolve based on research findings, user feedback, and technical feasibility identified throughout the development process.

1. Main Functionalities

The Meal Planner App will provide the following core functionalities, designed to support elderly users in managing their nutrition safely, efficiently, and independently:

- **Personalized weekly meal planning** structured by daily meal slots e.g. breakfast, lunch, dinner, based on professional nutritional guidelines and individual user profiles, including dietary restrictions, allergies, medical considerations, cultural preferences, and budget constraints.
- **Pre-approved recipe recommendations** that comply with nutritionist-defined rules, ensuring meals are both healthy and suitable for the user.
- **Clear, step-by-step meal preparation guidance** tailored for elderly users, focusing on simplicity, clarity, and ease of understanding.
- **Meal tracking and feedback**, allowing users to monitor completed meals and maintain adherence to daily and weekly meal plans.
- **Smart shopping list generation**, automatically created from selected meals and ingredients, with optional barcode scanning support to record used items and refine shopping list.
- **Caregiver and family coordination support**, enabling shared responsibility for shopping and meal preparation where required.
- **Optional integration with shopping or grocery delivery services**, to further reduce the effort associated with food-related tasks.

All functionalities will be designed with accessibility, usability, and simplicity as primary concerns, ensuring the application remains intuitive for users with limited technical experience.

2. Non-Functional Characteristics

The following non-functional characteristics define the quality attributes of the Meal Planner App and guide its overall design and implementation:

- **Usability**: The application will prioritize ease of use through a simple navigation structure, clear language, and minimal cognitive load [1], ensuring it is suitable for elderly users with varying levels of technical experience.
- **Accessibility**: The user interface will follow established accessibility principles, including readable typography, sufficient contrast, and support for users with reduced vision or motor abilities.

- **Reliability**: The system should behave consistently and predictably, ensuring meal plans, recipes, and shopping lists are available when needed without frequent errors or failures.
- **Performance**: The application should respond quickly to user actions, with minimal loading times to avoid frustration and confusion for elderly users.
- **Security and Privacy**: User data, including dietary preferences and health-related information, will be handled securely and with respect for user privacy.
- **Maintainability**: The system will be designed to allow future enhancements and changes, supporting the agile development approach adopted for the project.

3. Target Users and Stakeholders

The Meal Planner App is intended to support multiple user groups, each with distinct needs and responsibilities:

- **Primary Users – Elderly Individuals**
  Elderly users who wish to maintain a healthy diet and independent lifestyle.[3]. These users may have specific dietary requirements, medical constraints, or limited technical experience, making simplicity and accessibility critical.
- **Secondary Users – Caregivers and Family Members**
  Family members or caregivers who assist elderly users with meal planning, shopping, or meal preparation. The application aims to support coordination and shared responsibility between caregivers and elderly users.
- **Professional Stakeholders – Nutritionists and Healthcare Advisors**
  Nutrition professionals who define dietary guidelines, restrictions, and recommendations that inform meal planning rules within the system.
- **Service Stakeholders – Grocery and Delivery Providers**
  External services that may support food shopping or delivery, helping to reduce the physical burden of food-related tasks for elderly users.
- **Academic Stakeholders – Supervisors and Evaluators**
  Project supervisors and academic assessors who review the project deliverables, ensuring the system meets educational, technical, and documentation standards.

4. Platform

The Meal Planner App will be delivered as a **mobile-first web application**, ensuring broad accessibility across devices without requiring platform-specific installation. This approach

allows elderly users and caregivers to access the system using smartphones, tablets, or desktop browsers, while reducing technical barriers and maintenance overhead.

The platform choice prioritizes ease of access, rapid iteration, and cross-device compatibility, which aligns with the agile development approach adopted for the project. A web-based platform also enables straightforward deployment, updates, and feedback collection during iterative development.

Subject to project outcomes and future expansion, the platform may be extended to include a **native mobile application**, leveraging the same core functionality while offering enhanced device integration where beneficial.

5. Business Model Canvas

The Business Model Canvas below presents a structured and market-oriented view of the Meal Planner App. Although initiated as an academic project, the proposed model reflects a realistic digital health product with clear potential for future commercial deployment and scaling.

Meal Planner for Elderly				
Key Partners	**Key Activities**	**Value Proposition**	**Customer Relationships**	**Customer Segments**
cloud service providers	Shopping list generation	nutritionist-approved meal plans	Guided user interaction	Elderly individuals
Nutritionists	Nutrition rule management	Accessible and easy-to-use interface	Caregiver-supported usage	Family members and caregivers
Grocery and delivery services	Continuous usability and accessibility optimization	Support for independent living		Healthcare organisations (future)
Grocery retailers and food delivery service providers	Meal planning and recommendation	Reduced stress around meal planning and shopping		

	Key Resources		Channels	
	Dietary / Nutrition guidelines		Mobile-friendly web application	
	Curated / API-based recipe datasets		Caregiver access via shared accounts or permission-based access	
	Software platform			
	nutritional knowledge			
Cost Structure		Revenue Stream		
design and development costs		Freemium access		
system maintenance and hosting expenses		Subscription model		
Third-party service integrations		Partner integrations		
Data security, compliance, privacy management				

6. System Context

The Meal Planner App serves elderly users and caregivers by providing personalized meal plans, recipes, and shopping lists.

As shown in the diagram, the core system exchanges key information with four external entities:

- **Nutritionists/Healthcare Advisors** provide the **Nutritional Guidelines & Dietary Rules** that inform all meal recommendations.

- **Users (Elderly/Caregiver)** submit **Requests & Preferences** and receive **Personalized Meal Plans**.
- An **External Recipe Database** supplies detailed **Recipe Content & Nutritional Data**.
- **Grocery Retailers** receive **Formatted Shopping Lists/Orders** to facilitate food purchasing.

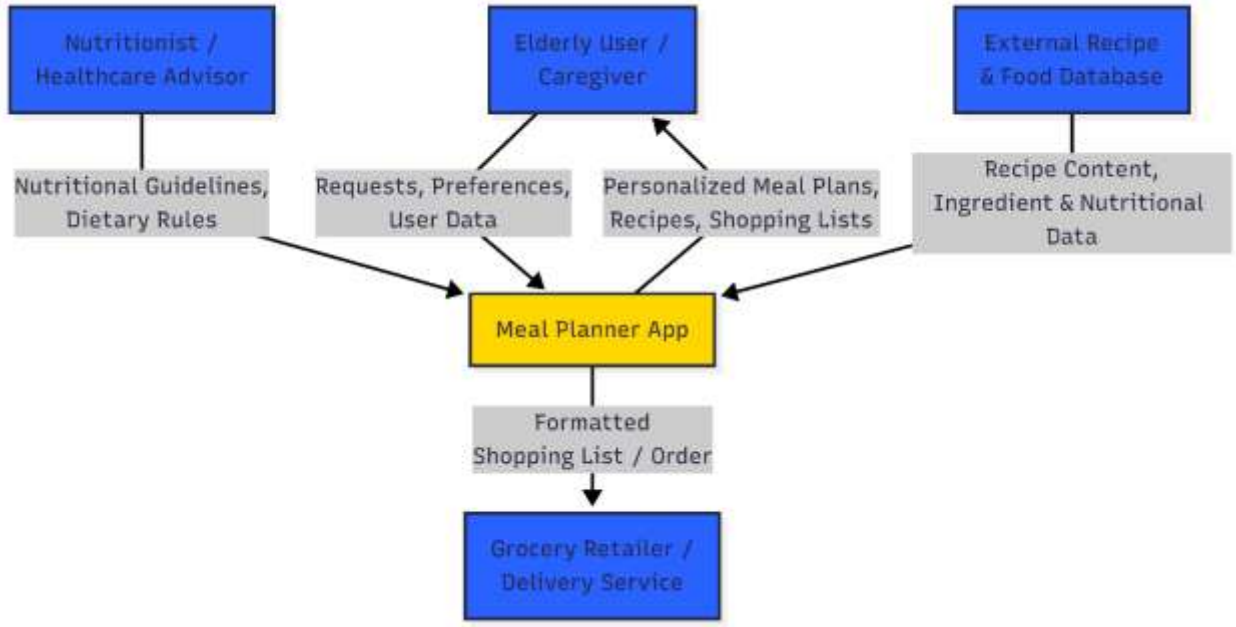

Fig.1. Context Diagram [Author's own]

7. Main risks

The following key risks have been identified at this stage of the project. These risks will be continuously reviewed and managed throughout the agile development process.

- **Usability risk for elderly users Risks**
  Designing an interface that is intuitive, accessible, and easy to use for elderly users with diverse cognitive, visuals, or motor abilities may be challenging[1][5].
- **Nutritional accuracy and liability Risks**
  Ensuring that meal recommendations correctly follow professional nutritional guidelines and do not conflict with medical conditions or medications is critical and requires careful validation [3].
- **Data privacy and security Risks**
  The handling of personal and health-related user data introduces privacy, security, and regulatory risks that must be managed appropriately.

- **External dependency Risks**
  Integration wit third-party services such as recipe data providers or grocery delivery platforms may introduce risks related to availability, reliability, cost, or changes in external APIs.

- **Scope creep Risks**
  The wide range of potential features may lead to uncontrolled scope expansion if priorities are not clearly defined and managed during each iteration

8. High-level iteration plan

Meal planner for elderly will be developed using an agile, iterative approach consisting of multiple development cycles.

Inception Phase

The inception phase focuses on defining project vision, identifying core requirements, assessing risks, and conducting initial research into elderly user needs, nutrition guidelines, and comparable systems.

Iteration 1 Core Functionality

The first iteration will be based on delivering core values of the application. This includes basic user interaction, user profile management, nutrition-rule based meal planning, and presentation of pre-approved recipes.

Iteration 2 Enhanced Features and Integration

The second iteration will be based on extending the system with additional features such as meal tracking, smart shopping list generation, and initial support for caregiver. Where appropriate, integration with external recipe data source or shopping services will be explored.

Iteration 3 Usability, Refinement, and Evaluation

The final iteration will be based on improving usability, accessibility, and overall use experience, particularly for elder users. Testing, evaluation, and refinement based on feedback will be prioritized, along with deployment readiness and final documentation

Bibliography

[1] Norman, D. A. (2013). *The Design of Everyday Things*. Revised and Expanded Edition. Basic Books. (**Accessed:** 20 January 2026)
Classic reference for usability, simplicity, and human-centred design.

[2] Sommerville, I. (2016). *Software Engineering*. 10th Edition. Pearson. (**Accessed:** 20 January 2026)
Standard reference for Agile development, system context, and non-functional requirements.

[3] National Institute on Aging (NIA) – Healthy Eating for Older Adults. (**Accessed:** 21 January 2026)
https://www.nia.nih.gov
Practical guidance on dietary needs of elderly users.

[4] Strategyzer – Business Model Canvas (**Accessed:** 21 January 2026)
https://www.strategyzer.com
Official source of the Business Model Canvas framework.

[5] W3C – Web Content Accessibility Guidelines (WCAG) 2.1 (**Accessed:** 22 January 2026)
https://www.w3.org/WAI/standards-guidelines/wcag/
Global standard for accessibility requirements.

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